Open Elective ECE

Reg. No | | | | | | | | | | |

[18EEOE09]

FIFTH SEMESTER B.Tech EXAMINATION, DECEMBER–2021

HYBRID ELECTRIC VEHICLES

Time: 3 Hours **Maximum Marks: 70**

Instructions:
i. Answer **FIVE** full Questions. Question number 1 and 2 are compulsory. Each question carries 14 marks.
ii. Missing data may be suitably assumed.

1. a. Draw a general layout of an EV and discuss the transmission characteristics. **14**

2. a. What are the battery parameters? Explain each briefly. **14**

3. a. List basic requirements of EV or EHV motors. **05**

b. Derive the Torque-Speed relationship equation of DC motor also draw the Torque-Speed characteristic curve of DC motor. **09**

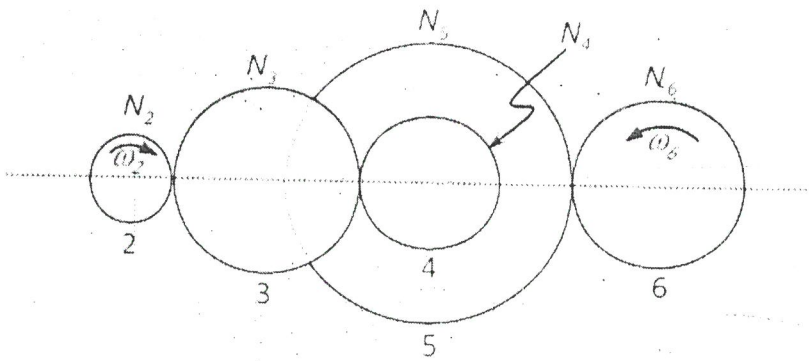
OR

4. a. Dissect the configuration and control of Switched reluctance motor **07**

b. Discuss in detail about the control of permanent magnet motor. **07**

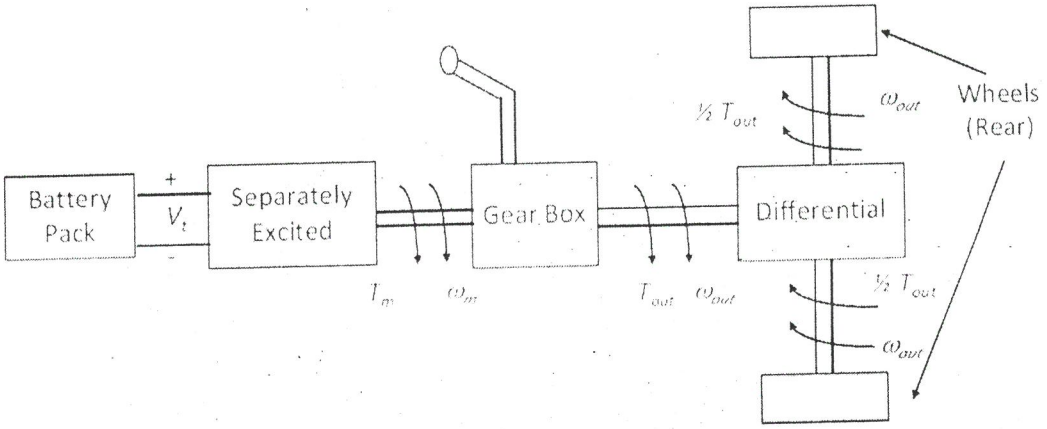
5. a. Derive the expression of Gear Ratio (GR) in terms of radius of the gears, angular speeds & number of teeth in the gears. **09**

b. What will be the Gear Ratio (GR) of the compound gear train shown in the figure below? **05**



OR

6. a. The powertrain of an electric vehicle is shown in the figure.
Given, V_t = 50 V, R_a = 0.7 Ω, $k\phi$ = 1.3, mg = 7,848 N.
Gear box gears: First GR_1 = 2, second GR_2 = 1, r_{wh} = wheel radius = 7 in. = 0.178 m. **14**
Find the vehicle maximum % gradability for each gear.



7.	a.	Compare serial & parallel HEVs. Also compare their advantages & disadvantages.	08
	b.	With a neat sketch, explain the configuration of Series hybrid electric drive train.	06

OR

8.	a.	With a neat sketch, explain the configuration of Series-Parallel combination HEV.	14